

CVM-X: CALIBRATING CONTINGENT VALUES WITH EXPERIMENTAL AUCTION MARKETS

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We design and implement a method, CVM-X, to calibrate hypothetical survey values using experimental auction markets. We test the procedure using consumer willingness-to-pay for irradiated/nonirradiated meat. Our results show that calibration factors for those who favor the irradiation process (0.67–0.69) are less severe than for those with an initial dislike of the process (0.55–0.59), suggesting that calibration may be commodity specific.

Key words: auctions, calibration, CVM-X, irradiated meat, valuation.

A central question regarding contingent valuation is whether values elicited from hypothetical surveys reflect the amounts individuals would actually pay for proposed programs (see, for example, Diamond and Hausman, and Hanemann). Economic values elicited from hypothetical questions can, under certain circumstances, provide useful information. For example, the panel commissioned by the National Oceanic and Atmospheric Administration (NOAA) concluded that willingness-to-pay values from contingent valuation studies could “produce estimates reliable enough to be the starting point of a judicial process of damage assessment” (NOAA 1993). NOAA, however, initially proposed a rule stipulating that compensable values estimated by CVM be deflated by 50% in the absence of evidence for an alternative calibration factor (NOAA 1994).¹ The proposed

deflation reflected evidence from several studies indicating an upward bias in CVM values.²

In this article, we develop a procedure, CVM-X, to measure and correct for differences that may exist between what people hypothetically say they would do and how they actually behave in a nonhypothetical situation. As proposed by Shogren (1993), the CVM-X method consists of four basic steps. First, we use a survey to elicit hypothetical values for the good in question. Second, we bring subsamples of the survey respondents into the laboratory and elicit real bids for the actual good in an incentive-compatible auction that employs real goods, real money, and repeated market experience. Third, we estimate a calibration function relating the auction market bids of the subsample to the hypothetical survey bids. Fourth, we use the estimated calibration function to adjust the values of the survey respondents who did not participate in the laboratory auction. Implicit in the CVM-X procedure is a test of validity, because we can directly compare hypothetical bids with those elicited under nonhypothetical

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¹ NOAA subsequently revised this proposal. The final version eliminates the need to calculate compensable values (NOAA 1996). Instead, valuation methods such as CVM would be used to determine only the appropriate scale of restorative actions rather than a monetary damage figure.

² The NOAA panel argued that an upward bias exists in the CVM and its recommendations stress a conservative design wherever possible. The panel also stressed the importance of validation: “External validation of the CVM method remains an important issue.” Similarly, McFadden suggests that research priority be given to “experiments that identify the biases that arise from CVM.” A number of studies have found that people overstate actual values when asked hypothetical questions. See, for example, Bishop and Heberlein; Bohm; Boyce and McCollum; Brookshire and Coursey; Cummings, Harrison, and Rutström; Dickie, Fisher, and Gerking; Duffield and Patterson; Kealy, Dovidio, and Rockel; Neill et al.; and Scip and Strand. Shogren (1990) and Shogren and Crocker observed that overstating one’s actual value declined with repeated market experience.

conditions in the laboratory. The expectation is that CVM-X can be a cost-effective tool that combines the advantages of CVM and experimental auction markets, thereby increasing the validity and accuracy of surveys while broadening the scope of nonmarket valuation in the laboratory. We are not, however, claiming that the results from one CVM-X study can be generalized to other market or nonmarket goods such as oil spills. Our results suggest that calibration seems commodity specific, and as such, CVM-X should proceed on a case-by-case basis until enough evidence reveals a systematic pattern leading to a general calibration function.

Calibration Procedures

Calibration can be interpreted as either the calibration of hypothetical and actual values for one good, or the use of a calibration function for one good to calibrate hypothetical values for a second good. Our CVM-X research takes the modest route—we focus on calibration for one good given hypothetical values from a CVM survey and real bids from an experimental auction market. Until we better understand if stated intentions in CVM surveys relate to real economic commitments, transferring bias across goods is premature. Our goal is to calibrate hypothetical and real commitments for the same good in different contexts, thereby explicitly accounting for the possibility that hypothetical bias may be good specific or context specific or both (also see List and Shogren). Before we turn to our CVM-X application, consider two recent attempts to calibrate hypothetical and real values across goods.

The cross-commodity bias function approach of Blackburn, Harrison, and Rutström rests on the presumption that bias for a good in one context is measurable and transferable to another good in another context. Blackburn, Harrison, and Rutström use discrete choice data on subjects who participated first in a hypothetical and then in a real valuation setting for one private good. A multinomial logit model is used to explain the pattern of responses (yes–yes, yes–no, no–no) and to determine whether the bias in hypothetical responses is systematically related to socioeconomic characteristics. Their evidence to support transferability of the bias function to a second private good is inconclusive given the relatively large standard errors of the esti-

mated coefficients (see Swallow's critique). In addition, their data do not permit estimation of value because they did not vary the price of the good. The use of one price, however, raises a more serious concern—the attempt to transfer a bias function is questionable because of the potential for price-dependent choice. For example, if one offers the good at a low price relative to value (e.g., gourmet chocolates for \$1 instead of \$5), one might observe a trivial discrepancy between hypothetical and real responses because a far greater proportion will actually purchase the good. It seems doubtful whether the bias function estimated for this context would be a useful predictor for a situation where a good (perhaps the same good) is offered at a higher price relative to value.

Harrison et al. calibrate open-ended values for wetlands protection using two bias functions: one to account for the downward bias due to free riding and the other to account for hypothetical bias. The free riding bias function is measured in a comparison of two real valuation situations for a nature calendar, one of which features public provision of the calendar. This bias function is used to calibrate both real and hypothetical bids for the preservation of local wetlands with the corrected bids used to estimate the hypothetical bias function. Both calibration functions, free riding followed by hypothetical, are then applied to bids elicited for national wetlands preservation. While intuitively appealing, the approach rests on little tested assumptions about transferability of bias functions between different contexts: real versus hypothetical values, private versus public goods. For example, the results are strongly influenced by the free riding correction but this correction rests on the implied assumption that incentives to free ride in a hypothetical valuation setting are similar to those in a real valuation setting. The hypothesis is testable—by eliciting hypothetical bids for the nature calendars with and without public provision. While this would seem to be a logical step in the transfer sequence, Harrison et al. choose instead to transfer free riding bias from real payments for private goods to hypothetical payments for public goods.³

³ An additional issue with Harrison et al. is that their calibration results could be path dependent. The free riding calibration is multiplicative—derived by treating the ratio of two bids as a function of respondents' characteristics. This in itself is problematic because it applies the same calibration factor to two individuals with the same

CVM-X Application

The Good: Trichinella Risk Reduction from Food Irradiation

The good we selected for our CVM-X application is the reduction in health risk from the parasite *Trichinella* achieved with food irradiation. Irradiation of pork has been approved by the United States Food and Drug Administration (FDA) and has been shown to reduce the viability of *Trichinella* organisms by greater than 99% (Brake et al.). The process has also been approved for a number of other products, and some market tests have indicated a high degree of consumer acceptability.⁴ But because irradiated foods are not yet widely available in the United States, many people are unfamiliar with the process—thereby giving it a feature common to many nonmarket goods.⁵ Lack of consumer familiarity and the ability to represent risk reduction in a tangible, private good that can be bought and sold on the auction block make irradiated pork a relevant and useful product to test the CVM-X method.

While scientists agree that food products properly treated with irradiation are completely safe, opponents of the process claim adverse health and environmental effects. Thus, consumers have been exposed to contradictory messages about the technology, resulting in some people having positive attitudes about the process and others having negative attitudes. We split the sample by attitudes, thereby allowing us to determine whether the potential bias in hypothetical versus nonhypothetical bids was symmetric across both groups (see Buhr et al.).

The Contingent Valuation Survey

A sample of 250 households in Story County, Iowa was randomly selected from telephone directory listings. An introductory letter ex-

socioeconomic characteristics regardless of the magnitude of their bids. Calibration for hypothetical bias accounts for the magnitude of the hypothetical bid with a calibration model of the general form: $WTP^h = \alpha WTP^r + f(X)$, where superscripts refer to predicted and hypothetical bids and X is the vector of characteristics. Reversing the order in which the calibrations are made, correcting first for hypothetical bias and then for free riding instead of vice versa, will result in a different calibrated bid, except under certain restrictions on parameters.

⁴ For example, an in-store study on consumer response to irradiated papayas in California found a high degree of acceptance. Irradiated papayas outsold regular papayas 10 to 1 (Bruhn and Noell).

⁵ Irradiated food products are currently available at only four retail outlets in the United States (three in Florida and one in Illinois).

Table 1. Summary of the CVM-X Survey

	Irradiated Preference	Non-irradiated Preference	No Preference	Total
Participated	58	22 ^a	2	82
Unavailable	39	8	1	48
Refused	34	10	0	44
Nonpork eater	6	1	1	8
Total	137	41	4	182

^a Includes the two subjects who refused to submit a hypothetical bid (see footnote 8).

plained the nature of the survey and indicated that the subject would be contacted by telephone to participate in a short interview. Professional interviewers subsequently contacted 208 subjects, from which were obtained 182 telephone interviews—a response rate of 87.5%.⁶ We sought as participants those most responsible for food shopping within the household.

The survey began with attitudinal questions related to diet and food safety. Demographic information was collected at the end. Following descriptions of trichinosis and food irradiation, participants were offered a hypothetical choice between a sandwich made with irradiated pork meat described as having a zero chance of causing trichinosis and a typical sandwich with nonirradiated pork described as having a 1 in 2.6 million chance of causing trichinosis.⁷ Table 1 shows that, of the 174 pork eaters in the sample, 131 (75%) indicated a preference for the irradiated pork sandwich. Other surveys on acceptability of irradiated meat products have found similar results (e.g., American Meat Institute Foundation).

Participants were asked the maximum they would be willing to pay to upgrade from their less-preferred sandwich to their sandwich of choice in an open-ended elicitation question.⁸

⁶ The survey instrument was developed with input from statisticians, veterinarians, and food scientists. After pretesting was conducted, we made minor changes to the descriptions of irradiation and trichinosis and reversed the order of two questions. A training session was conducted with professional interviewers prior to the pretest. The introductory letter, survey instrument, and experimental instructions are available from the authors on request.

⁷ Odds of *Trichinella* infection are based on Bennett et al.

⁸ The appropriate format for the elicitation question is itself an open question. Some argue that the closed-ended or discrete choice approach is more appropriate because it is easier for participants to understand, while others argue that the open-ended form is more appropriate because it provides more information by not dampening

Table 2. Selected Characteristics of Survey and Laboratory Participants

Variables	Irradiated Pork Preference		Nonirradiated Pork Preference	
	Survey (<i>N</i> = 131)	Laboratory Subsample (<i>N</i> = 58)	Survey (<i>N</i> = 38)	Laboratory Subsample (<i>N</i> = 20)
Age ^a	43.4	39.7	41.4	40.7
Gender (% male) ^b	0.32	0.33	0.26	0.20
Education ^c	5.1	5.3	5.1	5.3
Income ^d	3.5	3.5	3.8	4.1
Mean survey bid	\$0.61	\$0.58	\$0.58	\$0.67
(Variance)	(0.94)	(0.34)	(1.36)	(1.37)
Median survey bid	\$0.50	\$0.50	\$0.45	\$0.20
Zero bids	18	7	15	10
(%)	(0.14)	(0.12)	(0.39)	(0.50)

^a Age denotes actual age in years.

^b Gender denotes categorical variable (0–1): 0 if female; 1 if male.

^c Education denotes categorical variable (1–9): 1 = grade 8 or less; 2 = grades 9–11; 3 = high school graduate; 4 = some technical, trade, or business school; 5 = some college (but no degree); 6 = college graduate; 7 = some graduate work; 8 = master's degree; 9 = Ph.D., D.V.M., D.D.S., M.D., etc.

^d Income denotes categorical variable (1–8): 1 = less than \$10,000; 2 = \$10,000–19,999; 3 = \$20,000–29,999; 4 = \$30,000–39,999; 5 = \$40,000–49,999; 6 = \$50,000–69,999; 7 = \$70,000–100,000; 8 = more than \$100,000.

Among the 131 pork eaters preferring irradiated pork, the average hypothetical bid to upgrade from a nonirradiated pork sandwich to an irradiated pork sandwich was \$0.61. Of the 38 preferring nonirradiated, the average bid to upgrade to a nonirradiated sandwich was \$0.58. Two participants refused to submit a bid, with both stating that they would pay “a lot more” for the nonirradiated sandwich.⁹

At the end of the interview, participants who were pork eaters were asked if they would be interested in participating in a consumer economics experiment in return for a payment of \$30. Those confirming interest were asked to indicate a convenient time. The interviewers made no connection between the interview and the experiment. If asked, interviewers honestly informed subjects that they did not know anything about the experiments. Of the 174 eligible, 130 indicated they would be willing to participate; however, when contacted, 48 subjects were unavailable to participate at the scheduled times (table 1).

Table 2 provides the mean and median bids and some demographic characteristics for the

two primary groups and for the subgroups participating in experiments. Participation rates were similar between the two groups, with 58 of 131 (44%) preferring irradiated pork and 20 of 38 (53%) preferring nonirradiated pork.¹⁰ Because we wanted to include as many participants as possible in the experiments, the experimental subsample was not selected on the basis of demographic or socioeconomic characteristics. Table 2 shows, however, that our subsamples were representative of the larger groups.

The Experimental Auctions

Based on preferences revealed in the survey, participants were assigned to one of two treatments—the irradiated or the nonirradiated treatment.¹¹ Those preferring irradiated pork were assigned to the irradiated treatment; they were endowed with a nonirradiated pork sandwich and given the opportunity to bid for an irradiated pork sandwich. Similarly, those preferring nonirradiated pork were assigned to

the magnitude of subjects in the tail of the value distribution (see Mitchell and Carson). Recently, Cummings, Harrison, and Rutström found evidence that the discrete choice method tends to overstate actual willingness to pay.

⁹ Both subjects agreed to participate in the experiments. They were taken aside during the experiment and told that to continue, they had to give a bid. One gave a bid of \$1 and the other \$20. Bids from these two subjects are not included in any tables or statistical analyses.

¹⁰ As noted in Forsythe, Palfrey, and Plott, we are not above the “open problems that are being encountered in almost all experimental work where the costs of conducting experiments places a significant constraint on the number of observations” (p. 549).

¹¹ “Treatment” refers to the two main types of experiments—one involving bids for irradiated pork, the other involving bids for nonirradiated pork. “Subtreatments” refer to different auction mechanisms: second-price auction, random *n*th-price auction, and one-shot auction. “Group” or “experiment” refers to the 10 actual runs of the experiment.

the nonirradiated treatment, endowed with an irradiated pork sandwich and given the opportunity to bid for a nonirradiated pork sandwich.

The experimental auction procedures followed Shogren et al. (1994b). Each experiment had two stages. In stage one, participants learned about the auction mechanism by auctioning brandname candy bars. Over five trials, each participant submitted a sealed bid to exchange an endowed candy bar for an auctioned candy bar. Following each trial, the ID number(s) of the highest bidder(s) and the market price were displayed. After all five trials, we controlled for wealth effects by randomly selecting one of the five trials to be the binding trial that determined who would receive the auctioned candy bar. Participants were informed up front that only one of the five trials would be binding, and that each trial had an equal probability of being binding.

Stage two introduced the irradiated and nonirradiated pork sandwiches. Participants were informed that they would have to consume a sandwich (either their endowed, less preferred, sandwich or the auctioned, preferred sandwich) to complete the experiment and leave with their income. Each stage-two auction (except in group 10, which was a single-trial auction) consisted of 10 trials, each with a uniform chance of being randomly selected as the binding trial. As in stage one, the ID number(s) of the highest bidder(s) and the market price from the auction were publicly displayed following each trial.

At the beginning of stage two, participants were given a note reminding them of their hypothetical bid from the telephone survey. This reminder introduces the possibility of "anchoring" as subjects may be unwilling to admit to error or exaggeration in the survey by substantially revising that bid. This design element could bias results in favor of the CVM and toward more conservative calibration factors.¹² Most subjects (50 of 78), however, did change their bid between trials 1 and 2, with 29 subjects recording changes of \$0.25 or more.¹³

Before proceeding with the auction, participants were informed that their hypothetical survey bid would be regarded as their trial 1

auction bid. The highest bidder's ID number and the market price for trial 1 were posted before participants were asked to bid in trial 2. Between bidding for trials 1 and 2 the experiment moved participants from a hypothetical scenario with no market price information as in most CVM applications to a non-hypothetical situation in which they have some information about market prices as in most market situations. Following trial 5, participants were given a guided tour of an on-campus meat irradiation facility and a detailed description of the food irradiation process. Questions about irradiation were answered at that time. Subjects then returned to the auction room and resumed bidding for trials 6 through 10.

Ten experiments were conducted; seven in the irradiated treatment and three in the non-irradiated treatment. The number of participants per group ranged between 5 and 13. Of the seven groups in the irradiated treatment, four used the Vickrey second-price, sealed-bid auction mechanism with multiple trials. In the second-price auction, the highest bidder in the binding trial purchased the product at a price equal to the amount of the second-highest bid (Vickrey). One group in the irradiated treatment used a one-shot second-price auction. Participants in that group were allowed to revise their bids following the tour of the irradiation facility. The remaining two groups in the irradiated treatment and all three groups in the nonirradiated treatment used a variant of the multiple-trial Vickrey auction—a random n th-price auction. With the random n th-price mechanism, there was a uniform chance within each trial that the auction would be a second-, third-, fourth-, etc. highest bid auction. For example, if the binding trial had been randomly drawn to be a fourth-price auction, the three highest bidders would purchase the good at a market price equal to the fourth-highest bid (see Shogren et al. 1994a for details).

Because we are interested in exploring the phenomena of valuation with and without market experience in experimental auctions, we made a pragmatic choice to design most of our auctions with repeated market experience. The good we auctioned, irradiated pork, may eventually be widely bought and sold on the open market. If this product were offered commercially, individuals would have time to incorporate what others think about the product, and to review relevant information. We gain insight into the dynamic nature of value

¹² Reminding participants of their hypothetical bid could potentially impact our results. Testing for the effect of this reminder may provide interesting lines of future inquiry.

¹³ Only 16 subjects (10 in the irradiated treatment, 6 in the non-irradiated treatment) maintained the same bid throughout the survey and all auction trials.

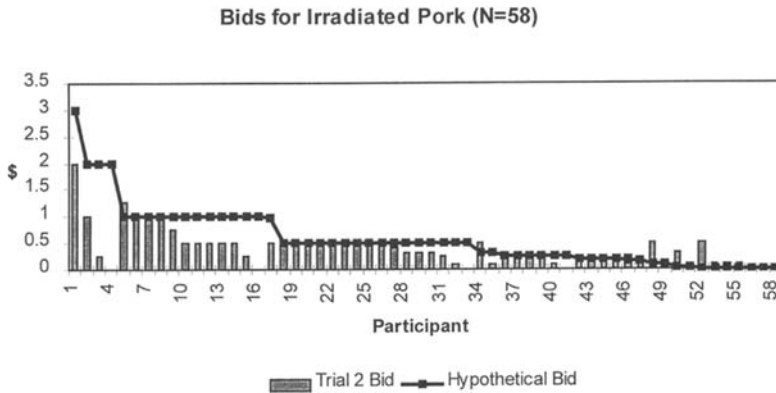


Figure 1. Comparison of hypothetical and auction bids for irradiated pork

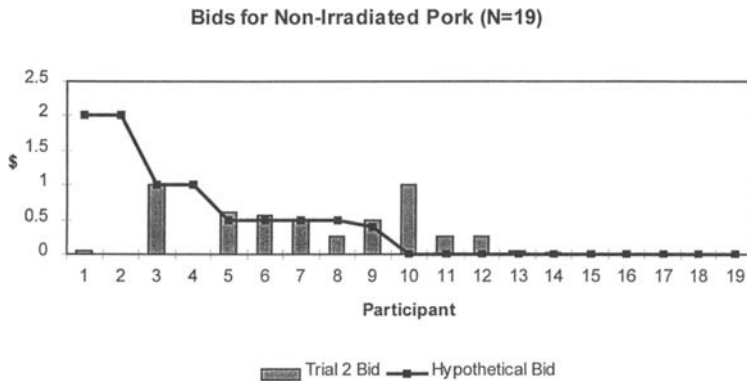


Figure 2. Comparison of hypothetical and auction bids for nonirradiated pork

elicitation by evaluating inexperienced and experienced bidding behavior in the initial trial and the later trials.

Anxiety may exist that our choice of repeated trials may have biased revealed values (see for example Rutström, or Harrison, Harstad, and Rutström). This raises solemn considerations. The long history of framing effects documented in the experimental and contingent valuation literature suggests that laboratory valuation exercises are also sensitive to design—different auctions generate distinct information flows that send unique signals to the bidders.¹⁴ Behavior depends on the context of choice. We recognize this, and it is why we chose the Vickrey-style auctions. In this auction, the dominant strategy is truth-telling—an incentive that holds unless the good has a substantial common value component such that the subjects' preferences are affiliated. If affiliation exists, new information

can bias bids upward (see Milgrom and Weber). As the results show in the following section, however, the evidence does not support affiliation: bids were unaffected by the choice of auction design and group size, as one would expect for the second-price auction without affiliation. While affiliation remains a conceptual concern, the results suggest it was not a significant issue in our treatments.

Comparison of Survey and Experimental Results

Figures 1 and 2 compare individual bids between the survey and trial 2 in both treatments.¹⁵ Table 3 shows the mean, median, and variance of the hypothetical, trial 2, and final bids for both treatments. The final bid is an

¹⁴ See, for example, Plott.

¹⁵ Figures 2 and 3 and table 3 include data for 19 of the 20 participants preferring nonirradiated pork. One outlying observation—that for an individual with a \$5 hypothetical bid and consistent zero bids in the auction—was eliminated.

Table 3. Comparison of Hypothetical and Experimental Auction Bids

	Irradiated Pork Preference (N = 58)			Nonirradiated Pork Preference (N = 19)		
	Survey	Trial 2 ^a	Final	Survey	Trial 2	Final
Mean bid	\$0.58	\$0.39	\$0.48	\$0.44	\$0.26	\$0.27
Variance	0.34	0.12	0.15	0.41	0.11	0.13
Median	0.50	0.30	0.42	0.00	0.05	0.02
Zero bids	7	6	6	10	7	8

^a Trial 2 is the first auction bid. The final market bid is an average over trials 8–10 except for the one-shot group, where it is represented by the revised bid.

average over trials 8, 9, and 10. Note that the variance of bids fell between the survey and the experimental auction. Figure 3 tracks the average bid in both treatments through all ten rounds of bidding. The change in the average bid between the survey and trial 2 in both treatments is similar in absolute terms. For nonirradiated pork, the average bid fell by \$0.18, from \$0.44 in the survey to \$0.26 in trial 2. The average bids for irradiated pork fell by \$0.19, from \$0.58 to \$0.39.

For nonirradiated pork, the median bid actually increased between the survey and the experiment due to fewer zero bids. A Wilcoxon signed-rank test did not reject equality of the hypothetical, trial 2, and final bid distributions in the nonirradiated treatment. But the correlation between hypothetical and trial 2 bids, as measured by the Spearman rank correlation coefficient, was low at 0.20 and insignificant ($p = 0.19$). Trial 2 and final bids were more highly correlated—the rank correlation coefficient was 0.39 ($p = 0.044$).

The median bid for irradiated pork fell from \$0.50 in the survey to \$0.30 in trial 2. The Wilcoxon test rejected equality of the distributions for hypothetical and trial 2 bids and also for trial 2 and final bids. Correlation between the hypothetical and trial 2 bids was greater than that for nonirradiated pork. The correlation coefficient was 0.58 and was significant ($p < 0.001$).

Figure 3 also shows the effect of the additional information about irradiation (e.g., the tour of the irradiation facility) provided to participants following trial 5. The average bid for irradiated pork increases while that for nonirradiated pork decreases, suggesting that on average the new information resulted in more favorable attitudes toward the irradiation process. In both cases the change in the average bid was insignificant. The change in bidding behavior as a result of the additional

information illustrates an advantage of the CVM-X procedure—because it is not always feasible to provide specific information (e.g., tours of the facility) in a large-scale survey, the impact of the information on a limited subsample can be translated through an appropriate calibration mechanism to the values from the larger sample.

Calibration

First, comparing simple average bids between the survey and the experiment provides rudimentary calibration factors for both goods. For irradiated pork this gives a calibration factor of 0.67 between the survey and trial 2 bids and a factor of 0.83 between survey and final bids. For nonirradiated pork, these calibration factors are 0.59 and 0.61. These calculations do not include the potential impact of demographic characteristics or factors related to how bids are elicited (e.g., groups, auction type) on the relationship between hypothetical and actual bids.

Tables 4 and 5 present alternative specifications of the calibration functions for irradiated and nonirradiated pork. Models A and B use the trial 2 bid as the dependent variable and models C and D use the final bid.¹⁶ Because seven of nineteen bids were zero, we used a tobit model for nonirradiated bids. We include both level and squared terms on the hypothetical bid as explanatory variables to approximate the unknown calibration function. Also included is a gender variable to test for differences in bid adjustment between males and females. Because the bid elicitation

¹⁶ Breusch-Pagan tests indicated heteroskedasticity in all models. We correct assuming multiplicative heteroskedasticity ($\sigma^2 = e^{\alpha Z}$), i.e., where the log of the error variance is a linear function of the specified exogenous variables in Z. In Z, we included the hypothetical bid and the revealed market price (see Greene, p. 409).

Table 4. Calibration Functions for Irradiated Pork ($N = 58$)

Variable	Model ^a			
	1A	1B	1C	1D
Constant	-0.14 (-0.10) ^b	0.09* (1.96)	-0.56 (-1.22)	0.24* (2.51)
Hypothetical bid	0.71* (4.41)	0.61* (3.94)	0.43* (2.20)	0.50* (2.19)
(Hypothetical bid) ²	-0.20 (-1.62)	-0.13 (-1.10)	-0.06 (-0.64)	-0.11 (-1.06)
Gender	0.05 (1.08)	0.08 (1.61)	-0.06 (-0.67)	0.05 (0.47)
Random n th	-0.01 (-0.14)		0.04 (0.27)	
One-shot	-0.11 (-1.12)		0.11 (0.61)	
Price 1 ^c	0.07 (0.63)		0.04 (0.15)	
Price 7			-0.07 (-0.13)	
Group size	0.01 (0.36)		0.09 (1.26)	
ECF ^d		0.69		0.86
OCF ^e		0.67		0.83
Log L	13.70	11.69	-13.01	-23.06

^a OLS models corrected for heteroskedasticity. For models 1A and 1B, the dependent variable is the bid reported in trial 2. For models 1C and 1D, the dependent variable is the average of bids reported in trials 8, 9, and 10.

^b The t -statistics are in parentheses; * denotes significance at the 95% level.

^c Price 1 and price 7 are the prices revealed prior to trials 2 and 8.

^d Estimated calibration factor.

^e Observed calibration factor, i.e., the ratio of the mean auction to mean survey bid for participants in the auction.

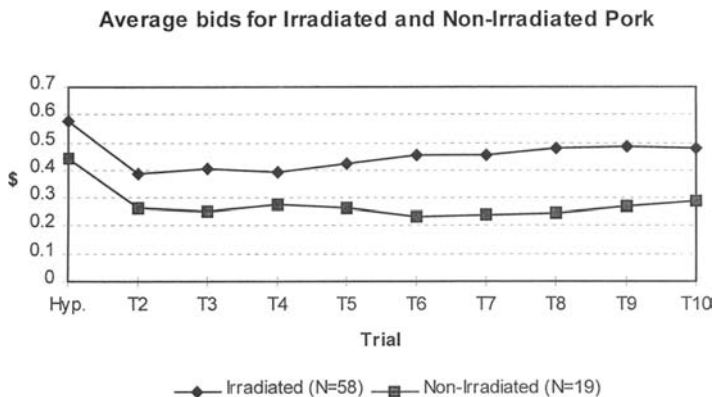


Figure 3. Average bids for irradiated and nonirradiated pork

procedure may affect the auction bids, we include dummy variables for the alternative auction mechanisms (random n th, one-shot) used in the irradiated treatment. Also included are the prices revealed to participants and the size of the group.¹⁷ Models B and D restrict the set of explanatory variables to those observable on nonparticipants.

In seven of eight models, the coefficient on the hypothetical bid is positive and significant. The squared term is generally insignificant for irradiated pork but significant for nonirradiated pork. In all models, Wald tests reject

restricting the coefficient on the hypothetical bid to either 1 (or 0.5 as implied by the initial NOAA proposal) at the 99% confidence level. The effect of gender is inconsistent and is statistically insignificant in all four of the unrestricted models.

The coefficients for alternative auction mechanisms in the irradiated treatment are both small and insignificant, suggesting consistency in bidding behavior across variations on the auction. The coefficient on revealed price (price 1) is positive but insignificant in all four instances. As expected, the coefficient on group size is also insignificant, suggesting no measurable relationship between bids and group size (see Davis and Holt, pp. 281–86).

¹⁷ Price 7 is not included in model 2C because of multicollinearity problems.

Table 5. Calibration Functions for Nonirradiated Pork (*N* = 19)

Variable	Model ^a			
	2A	2B	2C	2D
Constant	0.62 (0.81) ^b	-0.01 (-0.04)	1.56 (0.09)	-0.57* (-2.80)
Hypothetical bid	2.08* (9.77)	2.00* (3.25)	-1.49 (-1.16)	1.51* (5.71)
(Hypothetical bid) ²	-1.06* (-9.77)	-1.00* (-3.27)	0.87 (1.44)	-0.61* (-5.48)
Gender	-0.80 (-1.17)	-0.87* (-3.57)	0.14 (0.39)	0.56* (2.78)
Price 1 ^c	0.79 (0.40)		1.98 (0.21)	
Group size	-0.18 (-0.53)		-0.31 (-0.08)	
ECF ^d		0.55		0.28
OCF ^e		0.59		0.61
Log L	2.88	-5.08	3.96	3.53

^a Tobit models corrected for heteroskedasticity. For models 2A and 2B, the dependent variable is the bid reported in trial 2. For models 2C and 2D, the dependent variable is the average of bids reported in trials 8, 9, and 10.

^b The *t*-statistics are in parentheses; * denotes significance at the 95% level.

^c Price 1 is the price revealed prior to trials 2 and 8.

^d Estimated calibration factor.

^e Observed calibration factor, i.e., the ratio of the mean auction to mean survey bid for participants in the auction.

Calibration factors were estimated for both the trial 2 and final bids by applying the coefficients from models 1B, 1D, 2B, and 2D to the data available for survey respondents.¹⁸ The estimated calibration factor is the ratio of the mean predicted bid to the mean survey bid. The models for irradiated pork produce calibration factors similar to the rudimentary factors obtained by comparing average survey and auction bids for auction participants; 0.67 for the trial 2 bid versus 0.69 and 0.83 for the final bid versus 0.86. Even though the coefficient on gender in the calibration functions was insignificant, its impact on the calibration coefficient for males and females is worth noting. Model 1B produces a calibration factor of 0.87 for males versus 0.63 for females. Again, these factors are in close agreement with the rudimentary factors calculated for auction participants (0.87 for males and 0.58 for females).

The estimated calibration factors for the nonirradiated treatment are 0.55 for the trial 2 bid and 0.28 for the final bid. These factors were derived by applying the coefficients from models 2B and 2D to the survey data, then censoring the resulting predicted bids at zero. (Note that the parameter estimates in Table 5 represent the change in the latent bid.) The factor of 0.55 closely approximates the ratio of the mean trial 2 to survey bid for

participants (0.59), but the estimate of 0.28 for final bids is much lower than the corresponding ratio of observed means of 0.61. The lower calibration factors for the nonirradiated treatment suggest that bids from respondents who expressed an initial dislike of the irradiation process required more adjustment than those who were initially favorable, but this result must be treated with caution due to the limited number of participants in the nonirradiated treatment.

The possibility exists that some opponents of irradiation inflated their hypothetical bids to emphasize their opposition costlessly. This appeared to be the case for one participant whose bids were excluded in the calibration functions (see footnote 9). In the survey, this participant said she would pay \$20 hypothetically to avoid eating irradiated pork. When the laboratory auction began however, her bid immediately fell to \$1, to 5¢ in trial 5, and remained at 2¢ for trials 6 through 10. She consumed the irradiated pork.

Concluding Comments

The CVM-X procedure combines the attractive features of two valuation techniques—the broad sample sizes of surveys with the real consequences of experimental market auctions. Our goal is to increase the precision of field surveys and to increase the range of laboratory valuation exercises. Based on irradiated meat, which is a deliverable but unfamiliar good, our pilot CVM-X study suggests

¹⁸ One outlying bid was eliminated from the calibration calculation in each of the two treatments; a \$10 hypothetical bid in the irradiated treatment and a \$5 hypothetical bid in the nonirradiated treatment. Both bids fell outside the range of the estimated calibration functions.

that an upward bias in hypothetical bids exists but that the laboratory can be used to correct for this bias. By intentionally splitting the sample into proponents and opponents of irradiation, our results suggest that calibration may be commodity specific; the hypothetical bids for nonirradiated pork required greater adjustment. The notion that the calibration of hypothetical and actual bidding is good specific and context specific is supported by List and Shogren. In their field experiment on baseball cards, the average calibration factor ranged from 0.3 to 0.4—depending on the presence of other goods and intensity of market experience.

CVM practitioners are usually interested in public goods and programs, items generally lacking the immediate deliverability of an irradiated pork sandwich. If calibration is good specific and context specific as the current evidence suggests, further applications of CVM-X might be best restricted to more test-marketing of new private goods rather than moving immediately into the public goods arena. Until we first understand how the context of choice affects calibration for goods we can actually buy and sell, it is too early to claim progress in public good calibration. Future research needs to consider whether a private good can serve as a reasonable proxy for a public good, and whether a systematic method exists to cluster goods into classes of calibration functions.

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